

Stories in Space: A Model Tracks Story Arcs, Not Who Wrote Them

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Goodfire researchers mapped how one 4-bit [Llama 3.1 8B Instruct model](#) changed its reading of a story sentence by sentence. The [study](#) used 2,500 training stories and 500 held-out stories from [SimpleStories](#), a synthetic dataset with controlled styles. After each sentence, the model rated six emotions on a 0-to-10 scale while researchers also recorded its internal activity. Simple linear probes predicted the model's emotion and genre ratings with a root mean square

error of 0.09. Distances between concepts in the model’s reported ratings and internal activity correlated at $r = 0.92$ for emotions and $r = 0.89$ for genres. Steering activity across seven layers shifted those ratings, though pushing sadness also affected nearby concepts such as anger.

This gives interpretability researchers a way to watch a model’s idea of a story move over time, which could help trace where a summary or continuation starts reading the narrative differently. It does not reveal whether a person or AI wrote the text. The experiment never compared human-written and AI-written stories, and it tested one quantized model on synthetic data. The authors also [hand-picked three domains with six concepts each](#), treated each concept as applying to the whole story rather than one character, and reported no replication on another model family. The [Goodfire explanation](#) is best read as a map of one model’s changing state, not an authorship fingerprint.

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Sources:

- [Stories in Space paper](#)
- [Goodfire lab article](#)
- [SimpleStories dataset paper](#)
- [Llama 3.1 8B Instruct model card](#)

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